

Chapter 10: Derivatives of Multivariable Functions (§10.5 - 10.7)

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§10.5 The Chain Rule

In single variable calculus, if z depends on y and, in turn, y depends on x , The derivative of z w.r.t. x is given by

$$\frac{dz}{dx} = \frac{dz}{dy} \frac{dy}{dx}.$$

This most recent equation we call the Chain Rule.

Question

For $z = f(x, y)$, if x and y are functions of t , how to get $\frac{dz}{dt}$?

Preview Activity 10.5.1. Suppose you are driving around in the xy -plane in such a way that your position $\mathbf{r}(t)$ at time t is given by function

$$\mathbf{r}(t) = \langle x(t), y(t) \rangle = \langle 2 - t^2, t^3 + 1 \rangle.$$

The path taken is shown on the left of [Figure 10.5.1](#).

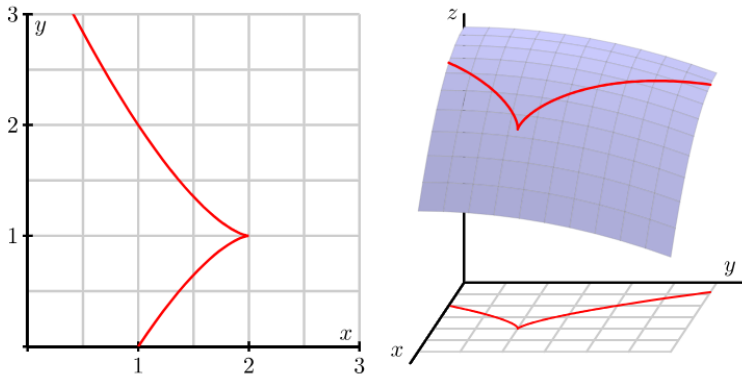


Figure 10.5.1. Left: Your position in the plane. Right: The corresponding temperature.

Suppose, furthermore, that the temperature at a point in the plane is given by

$$T(x, y) = 10 - \frac{1}{2}x^2 - \frac{1}{5}y^2,$$

and note that the surface generated by T is shown on the right of [Figure 10.5.1](#). Therefore, as time passes, your position $(x(t), y(t))$ changes, and, as your position changes, the temperature $T(x, y)$ also changes.

- The position function $\mathbf{r}$ provides a parameterization $x = x(t)$ and $y = y(t)$ of the position at time t . By substituting $x(t)$ for x and $y(t)$ for y in the formula for T , we can write $T = T(x(t), y(t))$ as a function of t . Make these substitutions to write T as a function of t and then use the Chain Rule from single variable calculus to find $\frac{dT}{dt}$. (Do not do any algebra to simplify the derivative, either before taking the derivative, nor after.)
- Now we want to understand how the result from part (a) can be obtained from T as a multivariable function. Recall from the previous section that small changes in x and y produce a change in T that is approximated by

$$\Delta T \approx T_x \Delta x + T_y \Delta y.$$

The Chain Rule tells us about the instantaneous rate of change of T , and this can be found as

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta T}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{T_x \Delta x + T_y \Delta y}{\Delta t}. \quad (10.5.1)$$

Use equation (10.5.1) to explain why the instantaneous rate of change of T that results from a change in t is

$$\frac{dT}{dt} = \frac{\partial T}{\partial x} \frac{dx}{dt} + \frac{\partial T}{\partial y} \frac{dy}{dt}. \quad (10.5.2)$$

- c. Using the original formulas for T , x , and y in the problem statement, calculate all of the derivatives in Equation (10.5.2) (with T_x and T_y in terms of x and y , and x' and y' in terms of t), and hence write the right-hand side of Equation (10.5.2) in terms of x , y , and t .
- d. Compare the results of parts (a) and (c). Write a couple of sentences that identify specifically how each term in (c) relates to a corresponding terms in (a). This connection between parts (a) and (c) provides a multivariable version of the Chain Rule.

The chain rule

The chain rule

Let $z = f(x, y)$, where f is a differentiable function of the independent variables x and y , and let x and y each be differentiable functions of an independent variable t . Then

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}.$$

From section 10.4 we have that when $z = f(x, y)$ differentiable,

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy.$$

Next, suppose that $x = x(t)$ and $y = y(t)$, so that

$$dx = \frac{dx}{dt} dt, \quad dy = \frac{dy}{dt} dt.$$

Thus we find that

$$dz = \frac{\partial f}{\partial x} \frac{dx}{dt} dt + \frac{\partial f}{\partial y} \frac{dy}{dt} dt.$$

Activity 10.5.2. In the following questions, we apply the Chain Rule in several different contexts.

- a. Suppose that we have a function z defined by $z(x, y) = x^2 + xy^3$. In addition, suppose that x and y are restricted to points that move around the plane by following a circle of radius 2 centered at the origin that is parameterized by

$$x(t) = 2 \cos(t), \text{ and } y(t) = 2 \sin(t).$$

- Use the Chain Rule to find the resulting instantaneous rate of change $\frac{dz}{dt}$.
- Substitute $x(t)$ for x and $y(t)$ for y in the rule for z to write z in terms of t and calculate $\frac{dz}{dt}$ directly. Compare to the result of part (i.).

- b. Suppose that the temperature on a metal plate is given by the function T with

$$T(x, y) = 100 - (x^2 + 4y^2),$$

where the temperature is measured in degrees Fahrenheit and x and y are each measured in feet.

- i. Find T_x and T_y . What are the units on these partial derivatives?
- ii. Suppose an ant is walking along the x -axis at the rate of 2 feet per minute toward the origin. When the ant is at the point $(2, 0)$, what is the instantaneous rate of change in the temperature dT/dt that the ant experiences. Include units on your response.
- iii. Suppose instead that the ant walks along an ellipse with $x = 6 \cos(t)$ and $y = 3 \sin(t)$, where t is measured in minutes. Find $\frac{dT}{dt}$ at $t = \pi/6$, $t = \pi/4$, and $t = \pi/3$. What does this seem to tell you about the path along which the ant is walking?

- c. Suppose that you are walking along a surface whose elevation is given by a function f . Furthermore, suppose that if you consider how your location corresponds to points in the xy -plane, you know that when you pass the point $(2, 1)$, your velocity vector is $\mathbf{v} = \langle -1, 2 \rangle$. If some contours of f are as shown in [Figure 10.5.2](#), estimate the rate of change df/dt when you pass through $(2, 1)$.

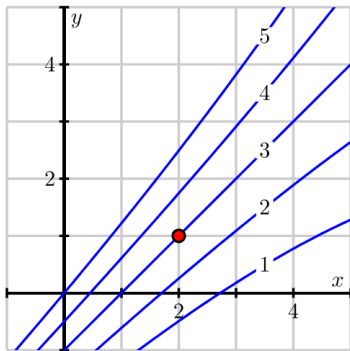
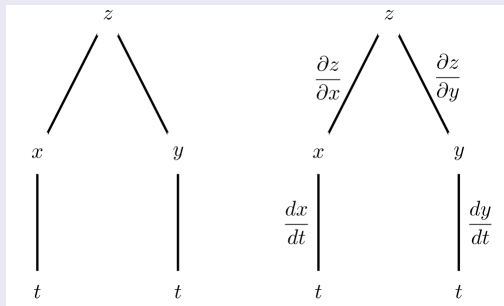


Figure 10.5.2. Some contours of f .

Multivariable cases

Tree diagrams

In more complicated cases (x and y depends on more than one quantity, or z depends on more than two variables), a tree diagram can be useful.



Activity 10.5.3.

- a. [Figure 10.5.4](#) shows the tree diagram we construct when (a) z depends on w , x , and y , (b) w , x , and y each depend on u and v , and (c) u and v depend on t .

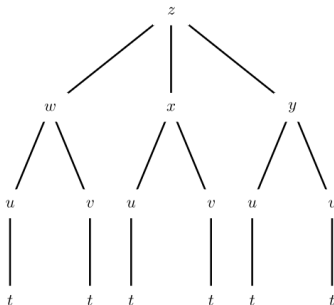


Figure 10.5.4. Three levels of dependencies

- Label the edges with the appropriate derivatives.
- Use the Chain Rule to write $\frac{dz}{dt}$.

b. Suppose that $z = x^2 - 2xy^2$ and that

$$x = r \cos(\theta)$$

$$y = r \sin(\theta).$$

- i. Construct a tree diagram representing the dependencies of z on x and y and x and y on r and θ .
- ii. Use the tree diagram to find $\frac{\partial z}{\partial r}$.
- iii. Now suppose that $r = 3$ and $\theta = \pi/6$. Find the values of x and y that correspond to these given values of r and θ , and then use the Chain Rule to find the value of the partial derivative $\frac{\partial z}{\partial \theta} \big|_{(3, \frac{\pi}{6})}$.

The following chain rule can be derived from our tree diagram.

The chain rule (case 2)

Suppose that $z = f(x, y)$ is differentiable for x, y , where $x = x(s, t)$ and $y = y(s, t)$ are differentiable functions of s and t . Then

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial s}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}, \quad \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial t}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}.$$

And it can be generalized.

The Chain rule (general version)

Suppose that f is a differentiable function of n variables $x_1, \dots, x_n$ and each x_j is a differentiable function of the m variables $t_1, t_2, \dots, t_m$ and

$$\frac{\partial f}{\partial t_i} = \sum_{k=1}^n \frac{\partial f}{\partial x_k} \frac{\partial x_k}{\partial t_i}$$

Example

If $z = f(x, y)$ has continuous second order partial derivatives and $x = r^2 + s^2$ and $y = 2rs$, find (a) $\partial z / \partial r$ and (b) $\partial^2 z / \partial r^2$.

Implicit Differentiation

Suppose that an equation of the form $F(x, y) = 0$ defines y implicitly as a differentiable function of x , that is, $y = f(x)$, where $F(x, f(x)) = 0$ for all x in the domain of f .

Question

How to determine $\frac{dy}{dx}$?

Suppose that an equation of the form $F(x, y, z) = 0$ defines z implicitly as a differentiable function of x and y , that is, $z = f(x, y)$, where $F(x, y, f(x, y)) = 0$ for all (x, y) in the domain of f .

Question

How to determine $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$?

The **Implicit Function Theorem** gives conditions under which $F(x, y) = 0$ defines y implicitly as a function of x . In our Calc III we do not need to consider the existence of implicit function but always assume that the given $F(x, y)$ defines y implicitly as a function of x .

determine $\frac{dy}{dx}$ from $F(x, y) = 0$

By applying the chain rule to differentiate both sides of the equation $F(x, y) = 0$ w.r.t. x , we obtain

$$\frac{\partial F}{\partial x} \frac{dx}{dx} + \frac{\partial F}{\partial y} \frac{dy}{dx} = 0 \implies \boxed{\frac{dy}{dx} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}} = -\frac{F_x}{F_y}}.$$

Note that we need the condition $F_y \neq 0$.

determine $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ from $F(x, y, z) = 0$

If F and f are differentiable (in this course they are!), by applying the chain rule to differentiate the equation $F(x, y, z) = 0$ as follows

$$\left\{ \begin{array}{l} \frac{\partial F}{\partial x} \frac{\partial x}{\partial x} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x} = 0 \implies \boxed{\frac{\partial z}{\partial x} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial z}} = -\frac{F_x}{F_z}} \\ \frac{\partial F}{\partial x} \frac{\partial x}{\partial y} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial y} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial y} = 0 \implies \boxed{\frac{\partial z}{\partial y} = -\frac{\frac{\partial F}{\partial y}}{\frac{\partial F}{\partial z}} = -\frac{F_y}{F_z}} \end{array} \right.$$

Note that we need the condition $F_z \neq 0$.

Examples

- Find y' if $x^3 + y^3 = 6xy$
- Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if $x^3 + y^3 + z^3 + 6xyz + 4 = 0$.

§10.6 Directional Derivatives and the Gradient

Recall that if $z = f(x, y)$, then f_x and f_y are defined as

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h}$$

$$f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y + h) - f(x, y)}{h}$$

Question

The partial derivatives of a function $f(x, y)$ tell us the rate of change of f in the direction of the coordinate axes (in the directions of the unit vectors $\mathbf{i}$ and $\mathbf{j}$). How can we measure the rate of change of f in other directions $\mathbf{u} = \langle a, b \rangle$?

Preview Activity 10.6.1. Let's consider the function f defined by

$$f(x, y) = 30 - x^2 - \frac{1}{2}y^2,$$

and suppose that f measures the temperature, in degrees Celsius, at a given point in the plane, where x and y are measured in feet. Assume that the positive x -axis points due east, while the positive y -axis points due north. A contour plot of f is shown in [Figure 10.6.1](#)

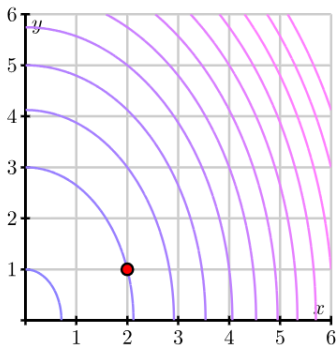
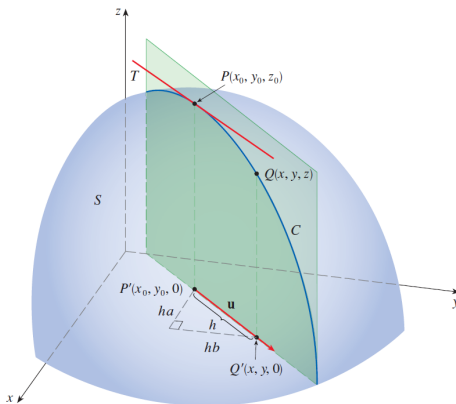


Figure 10.6.1. A contour plot of $f(x, y) = 30 - x^2 - \frac{1}{2}y^2$.

- a. Suppose that a person is walking due east, and thus parallel to the x -axis. At what instantaneous rate is the temperature changing with respect to x at the moment the walker passes the point $(2, 1)$? What are the units on this rate of change?
- b. Next, determine the instantaneous rate of change of temperature with respect to distance at the point $(2, 1)$ if the person is instead walking due north. Again, include units on your result.
- c. Now, rather than walking due east or due north, let's suppose that the person is walking with velocity given by the vector $\mathbf{v} = \langle 3, 4 \rangle$, where time is measured in seconds. Note that the person's speed is thus $|\mathbf{v}| = 5$ feet per second. Find parametric equations for the person's path; that is, parameterize the line through $(2, 1)$ using the direction vector $\mathbf{v} = \langle 3, 4 \rangle$. Let $x(t)$ denote the x -coordinate of the line, and $y(t)$ its y -coordinate. Make sure your parameterization places the walker at the point $(2, 1)$ when $t = 0$.
- d. With the parameterization in (c), we can now view the temperature f as not only a function of x and y , but also of time, t . Hence, use the chain rule to determine the value of $\left. \frac{df}{dt} \right|_{t=0}$. What are the units on your answer? What is the practical meaning of this result?

Directional derivatives

We now wish to find the rate of change of z at (x_0, y_0) in the direction of an arbitrary unit vector $\mathbf{u} = \langle a, b \rangle$. To do this we consider the surface S with the equation $z = f(x, y)$ and we let $z_0 = f(x_0, y_0)$. The vertical plane that passes through P in the direction of $\mathbf{u}$ intersects S in a curve C .



The slope of the tangent line T to C at the point P is the rate of change of z in the direction of $\mathbf{u}$.

If $Q(x, y, z)$ is another point on C , and $\langle x, y \rangle = \langle x_0, y_0 \rangle + h\mathbf{u}$ ($= \overrightarrow{P'Q'}$ in the figure). Then the increment of z is given by

$$\Delta z = f(x, y) - f(x_0, y_0) = f(x_0 + ha, y_0 + hb) - f(x_0, y_0).$$

And the limit $\lim_{h \rightarrow 0} \frac{\Delta z}{h}$ is the rate of change of z in the direction of $\mathbf{u}$, which is called the directional derivative of f in the direction of $\mathbf{u}$.

Definition: directional derivative

The derivative of f at the point (x, y) in the direction of the unit vector $\mathbf{u} = \langle a, b \rangle$ is denoted $D_{\mathbf{u}}f(x, y)$ and is given by

$$D_{\mathbf{u}}f(x, y) = \lim_{h \rightarrow 0} \frac{f(x + ha, y + hb) - f(x, y)}{h}$$

for those values of x and y for which the limit exists.

- $D_{\mathbf{u}}f(x_0, y_0)$ tells us the instantaneous rate at which f changes at (x_0, y_0) per unit increase in the direction of the vector $\mathbf{u}$.
- In addition, the quantity $D_{\mathbf{u}}f(x_0, y_0)$ tells us the slope of the line tangent to the surface in the direction of $\mathbf{u}$ at the point $(x_0, y_0, f(x_0, y_0))$.

Computing the Directional Derivative

Calculating a directional derivative

Given a differentiable function $f(x, y)$ and a unit vector $\mathbf{u} = \langle a, b \rangle$, we may compute $D_{\mathbf{u}}f(x, y)$ by

$$D_{\mathbf{u}}f(x, y) = f_x(x, y)a + f_y(x, y)b.$$

Moreover, if $\mathbf{u} = \langle \cos \theta, \sin \theta \rangle$, then the formula becomes

$$D_{\mathbf{u}}f(x, y) = f_x(x, y) \cos \theta + f_y(x, y) \sin \theta.$$

Proof.

For any (x_0, y_0) , define $g(h) = f(x_0 + ha, y_0 + hb)$. By the definition of derivative we have

$$g'(0) = \lim_{h \rightarrow 0} \frac{g(h) - g(0)}{h} = \lim_{h \rightarrow 0} \frac{f(x_0 + ah, y_0 + bh) - f(x_0, y_0)}{h} = D_{\mathbf{u}}f(x_0, y_0).$$

On the other hand, we can write $g(h) = f(x, y)$, where $x = x_0 + ha$ and $y = y_0 + hb$, so the chain rule gives

$$g'(h) = \frac{\partial f}{\partial x} \frac{dx}{dh} + \frac{\partial f}{\partial y} \frac{dy}{dh} = f_x(x, y)a + f_y(x, y)b.$$

Now we put $h = 0$ and $(x, y) = (x_0, y_0)$, then we have

$$D_{\mathbf{u}}f(x_0, y_0) = f_x(x_0, y_0)a + f_y(x_0, y_0)b.$$



Activity 10.6.2. Let $f(x, y) = 3xy - x^2y^3$.

- Determine $f_x(x, y)$ and $f_y(x, y)$.
- Use Equation (10.6.3) to determine $D_{\mathbf{i}}f(x, y)$ and $D_{\mathbf{j}}f(x, y)$. What familiar function is $D_{\mathbf{i}}f$? What familiar function is $D_{\mathbf{j}}f$? (Recall that $\mathbf{i}$ is the unit vector in the positive x -direction and $\mathbf{j}$ is the unit vector in the positive y -direction.)
- Use Equation (10.6.3) to find the derivative of f in the direction of the vector $\mathbf{v} = \langle 2, 3 \rangle$ at the point $(1, -1)$. Remember that a unit direction vector is needed.

The Gradient

Via the Chain Rule, we have seen that for a given function $f = f(x, y)$, its instantaneous rate of change in the direction of a unit vector $\mathbf{u} = \langle a, b \rangle$ is given by

$$\begin{aligned} D_{\mathbf{u}}f(x, y) &= f_x(x, y)a + f_y(x, y)b \\ &= \boxed{\langle f_x(x, y), f_y(x, y) \rangle} \cdot \langle a, b \rangle = \boxed{\nabla f} \cdot \mathbf{u}. \end{aligned}$$

We call the vector $\langle f_x(x, y), f_y(x, y) \rangle$ the **gradient** of f and denote it ∇f (which is read “the gradient of f ”, “grad f ” or “del f ”).

The directional derivative and the gradient

Given a differentiable function $f = f(x, y)$ and a unit vector $\mathbf{u} = \langle a, b \rangle$, we may compute $D_{\mathbf{u}}f(x, y)$ by

$$D_{\mathbf{u}}f(x, y) = \nabla f(x, y) \cdot \mathbf{u}.$$

Activity 10.6.3. Let's consider the function f defined by $f(x, y) = x^2 - y^2$. Some contours for this function are shown in [Figure 10.6.3](#).

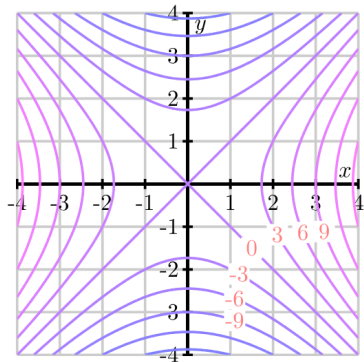


Figure 10.6.3. Contours of $f(x, y) = x^2 - y^2$.

- a. Find the gradient $\nabla f(x, y)$.
- b. For each of the following points (x_0, y_0) , evaluate the gradient $\nabla f(x_0, y_0)$ and sketch the gradient vector with its tail at (x_0, y_0) . Some of the vectors are too long to fit onto the plot, but we'd like to draw them to scale; to do so, scale each vector by a factor of $1/4$.
- $(x_0, y_0) = (2, 0)$
 - $(x_0, y_0) = (0, 2)$
 - $(x_0, y_0) = (2, 2)$
 - $(x_0, y_0) = (2, 1)$
 - $(x_0, y_0) = (-3, 2)$
 - $(x_0, y_0) = (-2, -4)$
 - $(x_0, y_0) = (0, 0)$
- c. What do you notice about the relationship between the gradient at (x_0, y_0) and the contour passing through that point?
- d. Does f increase or decrease in the direction of $\nabla f(x_0, y_0)$? Provide a justification for your response.

The Direction of the Gradient

If θ is the angle between $\nabla f(x_0, y_0)$ and $\mathbf{u}$ (where $\mathbf{u}$ is a unit vector), then

$$D_{\mathbf{u}}f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \mathbf{u} = |\nabla f(x_0, y_0)| |\mathbf{u}| \cos(\theta)$$

- When $\theta = \pi/2$, then $D_{\mathbf{u}}f(x_0, y_0) = 0$. This means that f is unchanging in this direction, and hence $\mathbf{u}$ must be tangent to the contour of f that passes through (x_0, y_0) . In other words, $\nabla f(x_0, y_0)$ is orthogonal to the contour through (x_0, y_0) .

Problem

Can you give an expression of $\mathbf{u}$ in this case? Why it is tangent to the contour $f(x, y) = f(x_0, y_0)$?

- When $\theta < \pi/2$ (acute), since $\cos(\theta) > 0$, it follows that $D_{\mathbf{u}}f(x_0, y_0) > 0$.
- When $\theta > \pi/2$ (obtuse), since $\cos(\theta) < 0$, it follows that $D_{\mathbf{u}}f(x_0, y_0) < 0$.

Finally, as we can see in the following activity, we may also use the gradient to determine the directions in which the function is increasing and decreasing most rapidly.

Activity 10.6.4. In this activity we investigate how the gradient is related to the directions of greatest increase and decrease of a function. Let f be a differentiable function and $\mathbf{u}$ a unit vector.

- a. Let θ be the angle between $\nabla f(x_0, y_0)$ and $\mathbf{u}$. Use the relationship between the dot product and the angle between two vectors to explain why

$$D_{\mathbf{u}}f(x_0, y_0) = |\langle f_x(x_0, y_0), f_y(x_0, y_0) \rangle| \cos(\theta). \quad (10.6.6)$$

- b. At the point (x_0, y_0) , the only quantity in Equation [\(10.6.6\)](#) that can change is θ (which determines the direction $\mathbf{u}$ of travel). Explain why $\theta = 0$ makes the quantity

$$|\langle f_x(x_0, y_0), f_y(x_0, y_0) \rangle| \cos(\theta)$$

as large as possible.

- c. When $\theta = 0$, in what direction does the unit vector $\mathbf{u}$ point relative to $\nabla f(x_0, y_0)$? Why? What does this tell us about the direction of greatest increase of f at the point (x_0, y_0) ?
- d. In what direction, relative to $\nabla f(x_0, y_0)$, does f decrease most rapidly at the point (x_0, y_0) ?
- e. State the unit vectors $\mathbf{u}$ and $\mathbf{v}$ (in terms of $\nabla f(x_0, y_0)$) that provide the directions of greatest increase and decrease for the function f at the point (x_0, y_0) . What important assumption must we make regarding $\nabla f(x_0, y_0)$ in order for these vectors to exist?

The Length of the Gradient

From Activity 10.6.4, we know that the direction in which a function increases most rapidly at a point (x_0, y_0) is the unit vector

$$\mathbf{u} = \frac{1}{|\nabla f(x_0, y_0)|} \nabla f(x_0, y_0) \quad (\nabla f(x_0, y_0) \neq \mathbf{0}).$$

Problem

In the direction of greatest increase for f at (x_0, y_0) , what is the value of the rate of increase $|D_{\mathbf{u}}f(x_0, y_0)|$?

Since

$$\begin{aligned} D_{\mathbf{u}}f(x_0, y_0) &= \nabla f(x_0, y_0) \cdot \left(\frac{1}{|\nabla f(x_0, y_0)|} \nabla f(x_0, y_0) \right) \\ &= \frac{1}{|\nabla f(x_0, y_0)|} (\nabla f(x_0, y_0) \cdot \nabla f(x_0, y_0)) \\ &= \frac{1}{|\nabla f(x_0, y_0)|} |\nabla f(x_0, y_0)|^2 = |\nabla f(x_0, y_0)|. \end{aligned}$$

We can summarize the results on the direction and length of the gradient.

Important facts about the gradient

Let f be a differentiable function and (x_0, y_0) a point for which $\nabla f(x_0, y_0) \neq \mathbf{0}$. Then $\nabla f(x_0, y_0)$ points in the direction of greatest increase of f at (x_0, y_0) , and the instantaneous rate of change of f in that direction is the length of the gradient vector. That is, if

$\mathbf{u} = \frac{1}{|\nabla f(x_0, y_0)|} \nabla f(x_0, y_0)$, then $\mathbf{u}$ is a unit vector in the direction of greatest increase of f at (x_0, y_0) , and $D_{\mathbf{u}}f(x_0, y_0) = |\nabla f(x_0, y_0)|$.

Activity 10.6.5. Consider the function f defined by $f(x, y) = -x + 2xy - y$.

- a. Find the gradient $\nabla f(1, 2)$ and sketch it on [Figure 10.6.5](#).

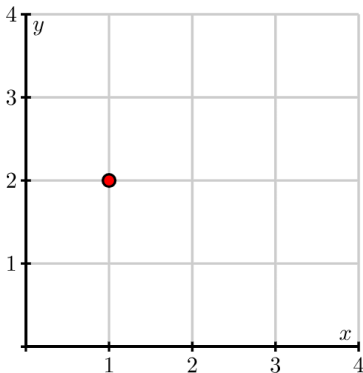


Figure 10.6.5. A plot for the gradient $\nabla f(1, 2)$.

- b. Sketch the unit vector $\mathbf{z} = \left\langle -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right\rangle$ on [Figure 10.6.5](#) with its tail at $(1, 2)$. Now find the directional derivative $D_{\mathbf{z}}f(1, 2)$.
- c. What is the slope of the graph of f in the direction $\mathbf{z}$? What does the sign of the directional derivative tell you?
- d. Consider the vector $\mathbf{v} = \langle 2, -1 \rangle$ and sketch $\mathbf{v}$ on [Figure 10.6.5](#) with its tail at $(1, 2)$. Find a unit vector $\mathbf{w}$ pointing in the same direction of $\mathbf{v}$. Without computing $D_{\mathbf{w}}f(1, 2)$, what do you know about the sign of this directional derivative? Now verify your observation by computing $D_{\mathbf{w}}f(1, 2)$.
- e. In which direction (that is, for what unit vector $\mathbf{u}$) is $D_{\mathbf{u}}f(1, 2)$ the greatest? What is the slope of the graph in this direction?
- f. Corresponding, in which direction is $D_{\mathbf{u}}f(1, 2)$ least? What is the slope of the graph in this direction?
- g. Sketch two unit vectors $\mathbf{u}$ for which $D_{\mathbf{u}}f(1, 2) = 0$ and then find component representations of these vectors.
- h. Suppose you are standing at the point $(3, 3)$. In which direction should you move to cause f to increase as rapidly as possible? At what rate does f increase in this direction?

Example

Consider a two-dimensional version of how a heat-seeking missile might work. Suppose that the temperature surrounding a fighter jet can be modeled by the function T defined by

$$T(x, y) = \frac{100}{1 + (x - 5)^2 + 4(y - 2.5)^2}$$

where (x, y) is a point in the plane of the fighter jet and $T(x, y)$ is measured in degrees Celsius.

Some contours and gradients ∇T are shown on the left in Figure 10.6.6.

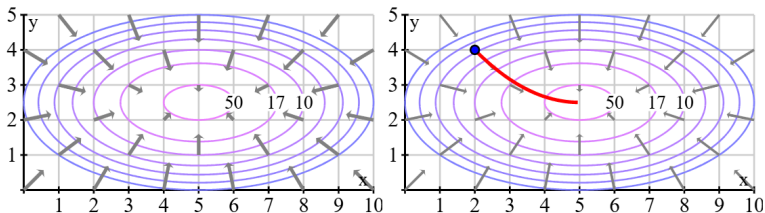


Figure 10.6.6. Contours and gradient for $T(x, y)$ and the missile's path.

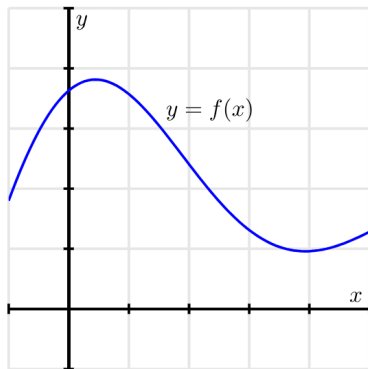
A heat-seeking missile will always travel in the direction in which the temperature increases most rapidly; that is, it will always travel in the direction of the gradient ∇T . If a missile is fired from the point $(2, 4)$, then its path will be that shown on the right in Figure 10.6.6.

Activity 10.6.6.

- The temperature $T(x, y)$ has its maximum value at the fighter jet's location. State the fighter jet's location and explain how [Figure 10.6.6](#) tells you this.
- Determine ∇T at the fighter jet's location and give a justification for your response.
- Suppose that a different function f has a local maximum value at (x_0, y_0) . Sketch the behavior of some possible contours near this point. What is $\nabla f(x_0, y_0)$? (Hint: What is the direction of greatest increase in f at the local maximum?)
- Suppose that a function g has a local minimum value at (x_0, y_0) . Sketch the behavior of some possible contours near this point. What is $\nabla g(x_0, y_0)$?
- If a function g has a local minimum at (x_0, y_0) , what is the direction of greatest increase of g at (x_0, y_0) ?

§10.7 Optimization

In single variable calculus, if a function f is everywhere differentiable, we know that the tangent line is horizontal at any point where f has a local maximum or minimum. Hence, one way that we seek extreme values of a given function is to first find where the derivative of the function is zero.



Preview Activity 10.7.1. Let $z = f(x, y)$ be a differentiable function, and suppose that at the point (x_0, y_0) , f achieves a local maximum. That is, the value of $f(x_0, y_0)$ is greater than the value of $f(x, y)$ for all (x, y) nearby (x_0, y_0) . You might find it helpful to sketch a rough picture of a possible function f that has this property.

- If we consider the trace given by holding $y = y_0$ constant, then the single-variable function defined by $f(x, y_0)$ must have a local maximum at x_0 . What does this say about the value of the partial derivative $f_x(x_0, y_0)$?
- In the same way, the trace given by holding $x = x_0$ constant has a local maximum at $y = y_0$. What does this say about the value of the partial derivative $f_y(x_0, y_0)$?
- What may we now conclude about the gradient $\nabla f(x_0, y_0)$ at the local maximum? How is this consistent with the statement " f increases most rapidly in the direction $\nabla f(x_0, y_0)$?"
- How will the tangent plane to the surface $z = f(x, y)$ appear at the point $(x_0, y_0, f(x_0, y_0))$?
- By first computing the partial derivatives, find any points at which $f(x, y) = 2x - x^2 - (y + 2)^2$ may have a local maximum.

Extrema and Critical Points

In single variable calculus, we use derivatives to identify local extremum of functions, now we can naturally extend the concept of local maxima and minima to several-variable functions.

Definition: local maximum/minimum & absolute max/min

Let f be a function of two variables x and y .

- The function f has a **local maximum** at a point (x_0, y_0) provided that $f(x, y) \leq f(x_0, y_0)$ for all points (x, y) near (x_0, y_0) . In this situation we say that $f(x_0, y_0)$ is a **local maximum value**.
- The function f has a **local minimum** at a point (x_0, y_0) provided that $f(x, y) \geq f(x_0, y_0)$ for all points (x, y) near (x_0, y_0) . In this situation we say that $f(x_0, y_0)$ is a **local minimum value**.
- An **absolute maximum point** is a point (x_0, y_0) for which $f(x, y) \leq f(x_0, y_0)$ for all points (x, y) in the domain of f . The value of f at an absolute maximum point is the **maximum value** of f .
- An **absolute minimum point** is a point (x_0, y_0) for which $f(x, y) \geq f(x_0, y_0)$ for all points (x, y) in the domain of f . The value of f at an absolute minimum point is the **minimum value** of f .

We use the term extremum point to refer to any point (x_0, y_0) at which f has a local maximum or minimum. In addition, the function value $f(x_0, y_0)$ at an extremum is called an extremal value.

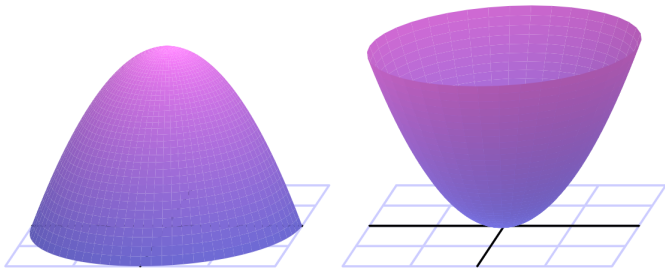


Figure 10.7.3. An absolute maximum and an absolute minimum

- In single-variable calculus, we saw that the extrema of a continuous function f always occur at critical points, values of x where f fails to be differentiable or where $f'(x) = 0$.
- Suppose that a continuous function $f(x, y)$ has an extremum at (x_0, y_0) . In this case
 - the trace $f(x, y_0)$ has an extremum at x_0 , which means that x_0 is a critical value of $f(x, y_0)$. Therefore, either $f_x(x_0, y_0)$ does not exist or $f_x(x_0, y_0) = 0$.
 - Similarly, the trace $f(x_0, y)$ has an extremum at y_0 , which means that y_0 is a critical value of $f(x_0, y)$. Therefore, either $f_y(x_0, y_0)$ does not exist or $f_y(x_0, y_0) = 0$.

Critical points

Critical points

A **critical point** of a function $f = f(x, y)$ is a point in the domain of f at which $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$, or such that one of $f_x(x_0, y_0)$ or $f_y(x_0, y_0)$ fails to exist.

Activity 10.7.2. Find the critical points of each of the following functions. Then, using appropriate technology, plot the graphs of the surfaces near each critical point and compare the graph to your work.

a. $f(x, y) = 2 + x^2 + y^2$

b. $f(x, y) = 2 + x^2 - y^2$

c. $f(x, y) = 2x - x^2 - \frac{1}{4}y^2$

d. $f(x, y) = |x| + |y|$

e. $f(x, y) = 2xy - 4x + 2y - 3.$

Classifying Critical Points: The Second Derivative Test

It is important to note that not every critical point leads to an extremum. For example

- $f(x) = x^3$ at $x_0 = 0$;
- $f(x, y) = x^2 - y^2$ at $(x_0, y_0) = (0, 0)$. We call $(0, 0)$ a saddle point because it is a local maximum on the trace $f(0, y)$ and a local minimum on the trace $f(x, 0)$.

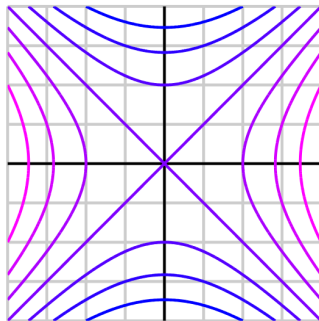
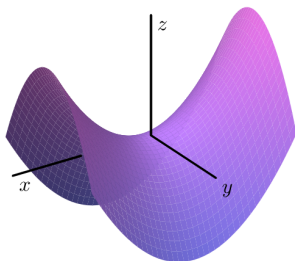


Figure 10.7.5. A saddle point.

The second derivative test: single variable functions

If x_0 is a critical point of a function f so that $f'(x_0) = 0$ and if $f''(x_0)$ exists, then

- if $f''(x_0) < 0$, x_0 is a local maximum;
- if $f''(x_0) > 0$, x_0 is a local minimum;
- if $f''(x_0) = 0$, this test yields no information ($f(x) = x^3, x^4, -x^4, \dots$)

Question

Do we have a similar second derivative test for classifying extreme values of functions of two variables?

Example

Consider the following three functions

$$f_1(x, y) = 4 - x^2 - y^2, \quad f_2(x, y) = x^2 + y^2, \quad f_3(x, y) = x^2 - y^2.$$

You can verify that each function has a critical point at the origin. You should check this.

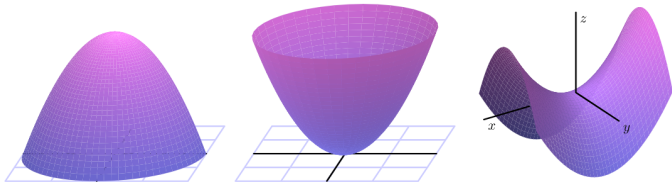


Figure 10.7.6. Three surfaces.

- a. The graphs of these three functions are shown in [Figure 10.7.6](#), with $z = 4 - x^2 - y^2$ at left, $z = x^2 + y^2$ in the middle, and $z = x^2 - y^2$ at right. Use the graphs to decide if a function has a local maximum, local minimum, saddle point, or none of the above at the origin.
- b. There is no single second derivative of a function of two variables, so we consider a quantity that combines the second order partial derivatives. Let $D = f_{xx}f_{yy} - f_{xy}^2$. Calculate D at the origin for each of the functions f_1 , f_2 , and f_3 . What difference do you notice between the values of D when a function has a maximum or minimum value at the origin versus when a function has a saddle point at the origin?
- c. Now consider the cases where $D > 0$. It is in these cases that a function has a local maximum or minimum at a point. What is necessary in these cases is to find a condition that will distinguish between a maximum and a minimum. In the cases where $D > 0$ at the origin, evaluate $f_{xx}(0, 0)$. What value does $f_{xx}(0, 0)$ have when f has a local maximum value at the origin? When f has a local minimum value at the origin? Explain why. (Hint: This should look very similar to the Second Derivative Test for functions of a single variable.) What would happen if we considered the values of $f_{yy}(0, 0)$ instead?

The second derivative test: two variable functions

The Second Derivative Test.

Suppose (x_0, y_0) is a critical point of the function f for which $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$. Let D be the quantity defined by

$$D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}(x_0, y_0)^2.$$

- If $D > 0$ and $f_{xx}(x_0, y_0) < 0$, then f has a local maximum at (x_0, y_0) .
- If $D > 0$ and $f_{xx}(x_0, y_0) > 0$, then f has a local minimum at (x_0, y_0) .
- If $D < 0$, then f has a saddle point at (x_0, y_0) .
- If $D = 0$, then this test yields no information about what happens at (x_0, y_0) .

The quantity D is called the *discriminant* of the function f at (x_0, y_0) .

Example

Find the critical points of the following functions and use the Second Derivative Test to classify the critical points.

❶ $f(x, y) = 3x^3 + y^2 - 9x + 4y$

❷ $f(x, y) = xy + \frac{2}{x} + \frac{4}{y}$

❸ $f(x, y) = x^3 + y^3 - 3xy$

Activity 10.7.5. While the quantity of a product demanded by consumers is often a function of the price of the product, the demand for a product may also depend on the price of other products. For instance, the demand for blue jeans at Old Navy may be affected not only by the price of the jeans themselves, but also by the price of khakis.

Suppose we have two goods whose respective prices are p_1 and p_2 . The demand for these goods, q_1 and q_2 , depend on the prices as

$$q_1 = 150 - 2p_1 - p_2 \quad (10.7.1)$$

$$q_2 = 200 - p_1 - 3p_2. \quad (10.7.2)$$

The seller would like to set the prices p_1 and p_2 in order to maximize revenue. We will assume that the seller meets the full demand for each product. Thus, if we let R be the revenue obtained by selling q_1 items of the first good at price p_1 per item and q_2 items of the second good at price p_2 per item, we have

$$R = p_1 q_1 + p_2 q_2.$$

We can then write the revenue as a function of just the two variables p_1 and p_2 by using Equations [\(10.7.1\)](#) and [\(10.7.2\)](#), giving us

$$\begin{aligned} R(p_1, p_2) &= p_1(150 - 2p_1 - p_2) + p_2(200 - p_1 - 3p_2) \\ &= 150p_1 + 200p_2 - 2p_1p_2 - 2p_1^2 - 3p_2^2. \end{aligned}$$

A graph of R as a function of p_1 and p_2 is shown in [Figure 10.7.7](#).

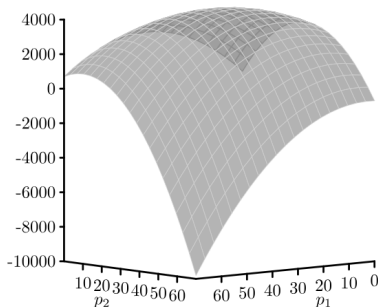


Figure 10.7.7. A revenue function.

- a Find all critical points of the revenue function, R . (Hint: You should obtain a system of two equations in two unknowns which can be solved by elimination or substitution.)
- b Apply the Second Derivative Test to determine the type of any critical point(s).
- c Where should the seller set the prices p_1 and p_2 to maximize the revenue?

Optimization on a Restricted Domain

We can always start with the single-variable calculus.

Extreme value theorem

A continuous function $f(x)$ on a closed interval $[a, b]$ always has both an absolute maximum and minimum on that interval, and that these absolute extremes must occur at either an endpoint or at a critical point.

From the extreme value theorem, we know that to find the absolute maximum and minimum, we determine the critical points in the interval and then evaluate the function at these **critical points** and **at the endpoints** of the interval.

Extreme value theorem: bivariate calculus

For functions of two variables, a **closed region** is a region that contains its boundary ($\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$), while a **bounded region** is one that does not stretch to infinity in any direction.

The Extreme Value Theorem

Let $f = f(x, y)$ be a continuous function on a closed and bounded region R . Then f has an absolute maximum and an absolute minimum in R .

The absolute extremes must occur at either a critical point in the interior of R or at a boundary point of R . We therefore must test both possibilities.

Example

Suppose the temperature T at each point on the circular plate $x^2 + y^2 \leq 1$ is given by

$$T(x, y) = 2x^2 + y^2 - y.$$

The domain $R = \{(x, y) : x^2 + y^2 \leq 1\}$ is a closed and bounded region, so the Extreme Value Theorem assures us that T has an absolute maximum and minimum on the plate. We will find the hottest and coldest points on the plate.

Questions

- 1 If the absolute maximum or minimum occurs inside the disk, it will be at a critical point, find all the critical points of T .
- 2 On the boundary of the disk $x^2 + y^2 = 1$, we have the following parametric representation

$$\begin{cases} x(t) = \cos(t) \\ y(t) = \sin(t) \end{cases}, \quad 0 \leq t \leq 2\pi.$$

By plugging $x(t), y(t)$ in $T(x, y)$, we get a single variable objective function (the function we want to optimize) $T(x(t), y(t))$. Now can you find the hottest and coldest points on the boundary of the disk?

Solution:

- 1 By solving $T_x = 0$ and $T_y = 0$ we have $x = 0$ and $y = 1/2$.
- 2 Since

$$T(x(t), y(t)) = 2 \cos^2(t) + \sin^2(t) - \sin(t) = \cos^2(t) - \sin(t) + 1,$$

we have $T'(t) = -2 \cos(t) \sin(t) - \cos(t) = -\cos(t)[2 \sin(t) + 1]$. Let $T'(t) = 0$, then

$$\cos(t) = 0 \text{ or } \sin(t) = -\frac{1}{2} \implies t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}.$$

The corresponding xy -coordinates are

$$(0, 1), (0, -1), \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right), \left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$$

We now have a list of candidates for the hottest and coldest points: the critical point in the interior of the disk and the critical points on the boundary. We find the hottest and coldest points by evaluating the temperature at each of these points, and find that

$$T\left(0, \frac{1}{2}\right) = -\frac{1}{4}, \quad T(0, 1) = 0, \quad T(0, -1) = 2$$
$$T\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right) = T\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right) = \frac{9}{4}.$$

So the maximum value of T on the disk $x^2 + y^2 \leq 1$ is $9/4$, which occurs at the two points $\left(\pm\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$ on the boundary, and the minimum value of T on the disk is $-1/4$ which occurs at the critical point $\left(0, \frac{1}{2}\right)$ in the interior of R .

Procedure for determining the max/min

Procedure for determining the absolute maximum and absolute minimum of a function on a closed and bounded domain

- Step 1: Find all critical points of the function in the interior of the domain.
- Step 2: Find all the critical points of the function on the boundary of the domain. Working on the boundary of the domain reduces this part of the problem to one or more single variable optimization problems. Note that there may be endpoints on portions of the boundary that need to be considered.
- Step 3: Evaluate the function at each of the points found in Steps 1 and 2.
- Step 4: The maximum value of the function is the largest value obtained in Step 3, and the minimum value of the function is the smallest value obtained in Step 3.

Activity 10.7.6. Let $f(x, y) = x^2 - 3y^2 - 4x + 6y$ with triangular domain R whose vertices are at $(0, 0)$, $(4, 0)$, and $(0, 4)$. The domain R and a graph of f on the domain appear in [Figure 10.7.10](#).

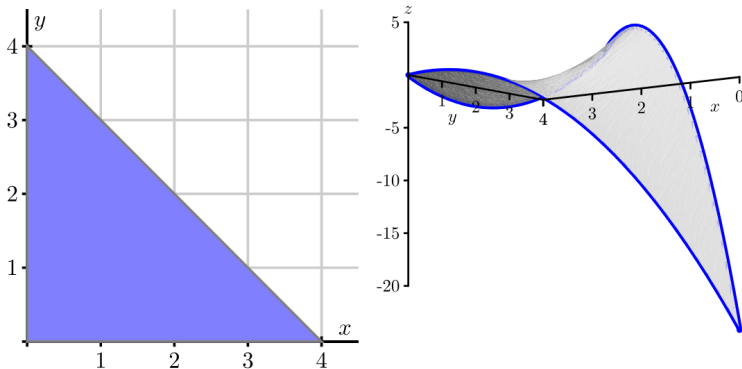


Figure 10.7.10. The domain of $f(x, y) = x^2 - 3y^2 - 4x + 6y$ and its graph.

- a. Find all of the critical points of f in R .
- b. Parameterize the horizontal leg of the triangular domain, and find the critical points of f on that leg. (Hint: You may need to consider endpoints.)
- c. Parameterize the vertical leg of the triangular domain, and find the critical points of f on that leg. (Hint: You may need to consider endpoints.)
- d. Parameterize the hypotenuse of the triangular domain, and find the critical points of f on the hypotenuse. (Hint: You may need to consider endpoints.)
- e. Find the absolute maximum and absolute minimum values of f on R .